

Organizational Conditions for AI-Driven Firm Performance

A Systematic Review of Empirical Evidence, 2015–2026

Bachelor's Thesis — interim draft for supervisor review

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Chapter 1

Introduction

This chapter follows. Research question and framing as approved in the proposal.

Chapter 2

Theoretical Background

2.1 Competitive advantage and firm performance

Competitive advantage is measured in 15 of the 67 studies reviewed here, and the remaining 52 report performance outcomes only. Barney separates the two constructs by what rivals are able to do: a firm holds a competitive advantage while it runs a value-creating strategy no rival, actual or potential, is running at the same time, and that advantage counts as sustained only once those rivals also cannot reproduce what the strategy yields (Barney, 1991, p. 102). Sustained in that sense says nothing about elapsed time. It asks whether the advantage is still there after rivals have given up trying to copy it (Barney, 1991, p. 102). This review therefore keeps the two apart throughout, because a firm can raise its margins without holding anything a rival is unable to copy.

Where both constructs appear, the separation is visible in the survey items. Chatterjee et al. ask about rivals directly: one of their competitive-advantage items states that implementing AI-based customer relationship management “will help an organization to win over its competitor which has not yet implemented AI-CRM for B2B relationship management” (Chatterjee et al., 2021, p. 215). Their performance items stay inside the firm and ask whether the same system improves “operational efficiency” and speeds up decisions (Chatterjee et al., 2021, p. 215).

Mehta et al. give competitive advantage a position between adoption and results: they test it as the channel through which AI adoption reaches performance, and predict a positive effect on performance in its own right (Mehta et al., 2025, p. 550).

Not every study that names competitive advantage builds its own instrument. Hossain et al. measure sustained competitive advantage with items carried over from earlier work (Hossain et al., 2022, p. 247). The construct label alone says little about what a study measured; the items decide.

2.2 Artificial intelligence as a general-purpose technology

Economists file AI under general-purpose technologies (GPTs), and that classification already predicts an uneven payoff. A GPT earns the label because it makes “new and complementary production methods” available, which raise output only over time (Czarnitzki et al., 2023, p. 188). Those complements are the expensive part. Before output moves, a firm has to redesign how work is organised, retrain staff, rework its software and accumulate managerial experience, and because none of that accumulation is booked as an asset, measured productivity dips before it climbs (Brynjolfsson et al., 2021, pp. 333–334). The lag is years, not quarters.

Two studies in this corpus show what that looks like at firm level. On German innovation-survey panel data, Czarnitzki et al. measure a clear productivity gain from AI use, for the yes/no measure and for the intensity measure alike (Czarnitzki et al., 2023, p. 201). Fontanelli et al. ask a different question of French firms and find that AI use raises the *volatility* of productivity growth; the effect sits with firms that buy AI from outside providers, and it weakens where more of the payroll is made up of ICT engineers and technicians (Fontanelli et al., 2025, pp. 1–2). Across the two results, the reading this review takes forward is that the AI investment is the constant and the complements are the variable.

2.3 The resource-based view and the imitability problem

Across the studies reviewed here, the resource-based view (RBV) is the recurring reason given for expecting AI to pay off. Barney sets four conditions on a resource before it can hold an advantage in place: it has to be valuable, rare among competitors and imperfectly imitable, and no substitute may do the same strategic job (Barney, 1991, pp. 105–106). Alwakid and Dahri restate the four conditions as the VRIN acronym (Alwakid & Dahri, 2025, p. 14). That is a shorthand Barney himself comes close to when he lists “rareness, inimitability, non-substitutability” as the characteristics that turn a firm attribute into a resource (Barney, 1991, p. 106). Hossain et al. put the same argument in terms of what a firm does with what it owns, and tie the long-run case to holdings rivals cannot match (Hossain et al., 2022, pp. 241–242).

Purchased AI fails that test, and the RBV literature says so plainly. Krakowski et al. note that AI as a technology resource costs almost nothing to reproduce and that little stands in the way of copying it, so the RBV expects whatever advantage it substitutes for to erode (Krakowski et al., 2023, p. 1426). Their own evidence relocates the advantage

rather than denying it: what resists imitation is the combination a firm builds, not either side of it (Krakowski et al., 2023, p. 1446).

Two consequences follow for this review. The RBV speaks to competitive advantage and not to performance, so a study reporting higher margins has not thereby shown that anything is defensible against rivals; the separation of the two constructs in Section 2.1 is a theoretical requirement, not only a coding convention. And if the resource itself is imitable, the variation worth studying lies in what surrounds it, which is why the conditions under which AI pays off are treated here as the object of analysis rather than as controls.

2.4 Automation and augmentation

The same investment can be deployed in two ways, and the strategy literature keeps them apart. Raisch and Krakowski define automation as machines taking a task over from a human, and augmentation as humans and machines working on a task together (Raisch & Krakowski, 2021, p. 192). Their argument against the popular advice to prefer augmentation is that the two cannot be held apart in practice. For any single task at any single moment the firm picks one approach, and that choice makes the two contradictory; widen the frame to several tasks over several years and each turns out to produce the other, which makes them interdependent (Raisch & Krakowski, 2021, p. 195).

Corpus studies pick the distinction up as a design choice with different returns. Zebec and Indihar Štemberger treat the two as the use cases through which AI adoption reaches performance, automation paying off in efficiency and augmentation in what people and machines manage jointly (Zebec & Indihar Štemberger, 2024, p. 7). Their model then returns no significant direct effect of process automation on process performance once tasks become complex, and the recommendation they draw from it is a boundary condition: automate what is routine, augment what is ambiguous (Zebec & Indihar Štemberger, 2024, pp. 16–17). Where the line falls is itself unstable. In the fragrance industry the machine took over part of the perfumers' problem solving and could not touch the part that depends on smelling a scent and judging how people will react to it (Krakowski et al., 2023, p. 1445).

2.5 Situated AI

Kemp starts from the puzzle the two previous sections produce: AI is available to everyone, so buying it should not yield an advantage, yet firms plainly differ in what they get out of it. His answer is situated AI, which he defines as AI hemmed in by the firm's own experience, structure and relationships (Kemp, 2024, p. 618). Three properties of the technology stand in the way, and the first of them is the imitability problem from

Section 2.3 under another name: AI is generic, explicit and myopic. Being generic puts AI on a level with general human capital, which competitors can acquire and put to work much as the adopter does (Kemp, 2024, p. 620).

What a firm does control is the situating. Kemp names three activities: grounding, which picks the experiences a firm's own AI learns from; bounding, which works on the experiences feeding a competitor's AI; and recasting, which keeps algorithms and the routines around them adjusting to each other (Kemp, 2024, pp. 618–619). The vocabulary is conditional throughout. None of the three is a property of the technology, and each is something a firm may or may not do.

Empirical work in the corpus arrives at the same place from the data side. Shi et al., studying Chinese listed firms, attribute the spread in outcomes to intangible complements, to how AI is governed, and to managerial capability that fits one setting only (Shi et al., 2025, p. 1). Kemp's paper is conceptual, so what it offers are propositions rather than findings. They are the propositions this review is positioned to examine, because the conditions Kemp derives are the conditions the coded studies report.

Chapter 3

Method

Interim summary. The full chapter — search documentation, screening protocol, coding procedure, and PRISMA flow diagram — follows in the next version of this draft.

The review follows the protocol announced in the proposal. The documented Scopus query was run on 17 July 2026 and returned 432 unique records: peer-reviewed journal articles in English, published 2015–2026, in the Scopus subject areas business, economics, and decision sciences. The query finds all three benchmark studies defined in advance (Babina et al., 2024; Krakowski et al., 2023; Lui et al., 2022).

For journal quality, the strict AJG 4*/4 threshold of the proposal was replaced by a broader, uniform inclusion rule: a record stays in when its journal is listed in the AJG 2024 with a rating of at least 2. One fixed ranking edition applies to all records, so the rule is transparent and non-selective. This filter reduced the corpus from 432 to 174 records; most of the excluded records appeared in journals not listed in the ranking.

Title and abstract screening tested four criteria: the study is empirical, works at the firm level, treats firms’ AI investment or adoption as its object, and measures a performance or competitive-advantage outcome. This step reduced the 174 records to 81. Reading the full texts removed another fourteen studies, twelve on content grounds and two that could not be retrieved. The final corpus holds 67 studies. Every screening decision is recorded with a reason code, and the full chain ($432 \rightarrow 174 \rightarrow 81 \rightarrow 67$) will be reported as a PRISMA-style flow diagram.

Each included study was coded into the state-of-the-art table announced in the proposal; an excerpt is attached in the Appendix (Tables A.1 and A.2). Following the supervisor’s feedback, the scheme separates superior firm performance from competitive advantage as distinct outcome constructs, each with its own measurement column.

Chapter 4

Results

In progress. Part one, the descriptive mapping of the corpus, is drafted; part two, the synthesis of conditions, is the next writing step.

Chapter 5

Discussion

This chapter follows.

Chapter 6

Conclusion and Future Outlook

This chapter follows.

References

- Alwakid, W., & Dahri, N. (2025). Harnessing AI capabilities and green entrepreneurial orientation for sustainable SME performance using SEM analysis approach. *Technology in Society*, 83. <https://doi.org/10.1016/j.techsoc.2025.103007>
- Barney, J. (1991). Firm resources and sustained competitive advantage [AJG 4*]. *Journal of Management*, 17(1), 99–120. <https://doi.org/10.1177/014920639101700108>
- Brynjolfsson, E., Rock, D., & Syverson, C. (2021). The productivity J-curve: How intangibles complement general purpose technologies [AJG 4]. *American Economic Journal: Macroeconomics*, 13(1), 333–372. <https://doi.org/10.1257/mac.20180386>
- Chatterjee, S., Rana, N., Tamilmani, K., & Sharma, A. (2021). The effect of AI-based CRM on organization performance and competitive advantage: An empirical analysis in the B2B context. *Industrial Marketing Management*, 97, 205–219. <https://doi.org/10.1016/j.indmarman.2021.07.013>
- Czarnitzki, D., Fernández, G., & Rammer, C. (2023). Artificial intelligence and firm-level productivity. *Journal of Economic Behavior and Organization*, 211, 188–205. <https://doi.org/10.1016/j.jebo.2023.05.008>
- Fontanelli, L., Guerini, M., Miniaci, R., & Secchi, A. (2025). Predictive AI and productivity growth dynamics: Evidence from French firms. *Journal of Economic Behavior and Organization*, 240. <https://doi.org/10.1016/j.jebo.2025.107336>
- Hossain, M., Agnihotri, R., Rushan, M., Rahman, M., & Sumi, S. (2022). Marketing analytics capability, artificial intelligence adoption, and firms’ competitive advantage: Evidence from the manufacturing industry. *Industrial Marketing Management*, 106, 240–255. <https://doi.org/10.1016/j.indmarman.2022.08.017>
- Kemp, A. (2024). Competitive advantage through artificial intelligence: Toward a theory of situated AI [AJG 4*]. *Academy of Management Review*, 49(3), 618–635. <https://doi.org/10.5465/amr.2020.0205>
- Krakowski, S., Luger, J., & Raisch, S. (2023). Artificial intelligence and the changing sources of competitive advantage [AJG 4*]. *Strategic Management Journal*, 44(6), 1425–1452. <https://doi.org/10.1002/smj.3387>
- Mehta, P., Chakraborty, D., Rana, N., Mishra, A., Khorana, S., & Kooli, K. (2025). AI-driven competitive advantage: The role of personality traits and organizational

- culture in key account management. *Journal of Business and Industrial Marketing*, 40, 543–569. <https://doi.org/10.1108/JBIM-03-2024-0205>
- Raisch, S., & Krakowski, S. (2021). Artificial intelligence and management: The automation–augmentation paradox [AJG 4*]. *Academy of Management Review*, 46(1), 192–210. <https://doi.org/10.5465/amr.2018.0072>
- Shi, Y., Wu, T., Qin, C., & Liu, B. (2025). The value-creating potential of AI: A multi-dimensional analysis of effects and mechanisms. *International Review of Financial Analysis*, 108. <https://doi.org/10.1016/j.irfa.2025.104694>
- Zebec, A., & Indihar Štemberger, M. (2024). Creating AI business value through BPM capabilities. *Business Process Management Journal*, 30, 1–26. <https://doi.org/10.1108/BPMJ-07-2023-0566>

Appendix

Coding table (excerpt)

An excerpt of the coding table: the 25 studies published in the highest-ranked journals (AJG 4* and 4, then 3; within a rating ordered by year) plus, at the end, the first three AJG 2 studies as examples of the lowest included rating tier — 28 of 67 coded studies in total. Split into study characteristics (Table A.1) and findings (Table A.2); study IDs refer to the full table.

Table A.1: Included studies: characteristics and design (excerpt: 28 of 67 studies — the 25 in the highest-ranked journals (AJG 4*, 4, 3) plus the first three AJG 2 studies; within a rating by year). Source: author’s compilation.

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S09	Mishra et al. (2022)	Journal of the Academy of Marketing Science (4*)	USA	19,000 firm-year observations, US-listed firms (COMPUSTAT + EDGAR 10-Ks), simultaneous equations; PSM robustness (1,801 treatment / 2,251 control firms)	panel econometrics	10-K text (share of AI keywords = AI focus)
S14	Krakowski et al. (2023)	Strategic Management Journal (4*)	multi-country	same chess players across conventional, centaur and engine tournaments (controlled competitive setting)	panel econometrics	AI (engine) adoption in competitive interaction
S17	Babina et al. (2024)	Journal of Financial Economics (4*)	USA	1,993 US Compustat firms; resume-based AI workforce measure (Cognism), robust to job-postings measure; IV = university AI graduate supply	panel econometrics	resume-based (share of AI-skilled workers as firm AI investment)
S07	Leoni et al. (2022)	International Journal of Operations and Production Management (4)	Italy	120 senior executives of manufacturing firms, PLS-SEM	survey-SEM	survey construct (AI adoption/use)
S45	Adams et al. (2026)	Journal of Monetary Economics (4)	USA	US firms matched to Lightcast job postings 2010Q1-2024Q1 (40,000+ job boards); AI-pricing jobs identified	panel econometrics	job postings (AI-skill + pricing keyword = AI pricing adoption)
S46	Agag et al. (2026)	Tourism Management (4)	UK	1,206 UK listed tourism/travel/hospitality firms, 7,539 firm-years 2018-2024; two-step system GMM	panel econometrics	AI human capital (AI-skilled workforce measure)
S52	Filieri et al. (2026)	Journal of Product Innovation Management (4)	China	Chinese listed manufacturers + WTO tariff records + customs data 2015-2022; IV analysis	panel econometrics	archival AI integration measure (tariff-induced)
S02	Baabdullah et al. (2021)	Industrial Marketing Management (3)	Saudi Arabia	392 B2B SMEs, cross-section, CB-SEM	survey-SEM	survey construct (acceptance of AI practices)
S03	Chatterjee et al. (2021)	Industrial Marketing Management (3)	India	349 managers from 16 BSE-listed firms (11 manufacturing, 5 service), response rate 51.6%, PLS-SEM	survey-SEM	survey construct (AI-CRM implementation)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S04	Chatterjee et al. (2022)	Journal of Business Research (3)	India	312 usable manager responses from 14 BSE-listed firms (10 large/4 SME; telecom, IT, automobile, retail, pharma), Jan-Feb 2020, RR 47.3%, PLS-SEM	survey-SEM	survey construct (adoption of AI-embedded CRM system)
S05	Hossain et al. (2022)	Industrial Marketing Management (3)	Bangladesh	257 usable manager responses (1,953 approached, 278 qualified), RMG export manufacturers, research-firm panel/Qualtrics, random sampling; time-lagged robustness subsample n=93	survey-SEM	survey construct (AI adoption on top of marketing analytics platform)
S06	Lee et al. (2022)	Technovation (3)	South Korea	160 high-tech ventures (from 1,248 ministry list, 300 responses), survey + audited financials	panel econometrics	survey (AI-adoption intensity) linked to financial records
S08	Lui et al. (2022)	Annals of Operations Research (3)	multi-market	119 AI investment announcements by 62 listed firms, Factiva search Jan 2015 - Feb 2019	event study	announcements (AI investment announcements)
S12	Czarnitzki et al. (2023)	Journal of Economic Behavior and Organization (3)	Germany	5,851 firms (409 AI users, ~7%), German CIS 2018 (ZEW); cross-section OLS/2SLS-IV + panel subset with IV-FE, entropy balancing	panel econometrics	survey (CIS AI adoption dummy + intensity of AI use in business processes)
S15	Samadhiya et al. (2023)	Industrial Marketing Management (3)	India	in-depth interviews with senior B2B managers + PLS-SEM on 142 responses	mixed	survey construct (AI-based partner relationship management implementation)
S16	Wu & Monfort (2023)	Psychology and Marketing (3)	Taiwan	278 food firms (CEO/marketing managers), Taiwan, survey Mar-Aug 2020, RR 33%; SEM + supplementary fsQCA	survey-SEM	survey construct (implementation of AI marketing strategy)
S18	Cannas et al. (2024)	International Journal of Production Research (3)	Italy	multiple case study: 6 companies, 17 AI implementation cases, semi-structured interviews	case study	case observation (AI applications in SCOR processes)
S20	Pham et al. (2024)	Journal of Business Research (3)	USA	941 AI hospital-year obs + 941 matched non-AI hospital-year obs (246 matched hospitals), 40 US states, 2000-2020	panel econometrics	adoption dummy (hospital AI adoption)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S21	Sullivan & Fosso (2024)	Journal of Business Research (3)	USA	two-wave survey, 107 US executives completing both waves (of 225 wave-1; 47.5% effective RR), panel-recruited IT+business executives	survey-SEM	survey constructs (AI-enabled automation, AI-enabled analytics, AI-enabled relational capabilities)
S24	Alam et al. (2025)	International Journal of Hospitality Management (3)	Malaysia	336 respondents, stratified purposive sampling, hospitality industry	survey-SEM	survey construct (AI adoption + technology readiness)
S26	Banna & Alam (2025)	International Journal of Entrepreneurial Behaviour and Research (3)	EU	unbalanced panel of 1,479 firms (~12,900 obs), 26 EU countries, 2012-2023; CGM multi-way clustering + 2SLS-IV	panel econometrics	AI-related venture-capital funding received (log), OECD AI Policy Observatory data (turning point USD 11.3M); robustness: AI job intensity, AI patents, industry-level IV
S27	Basnet et al. (2025)	International Review of Financial Analysis (3)	USA	US firms, 10-K filings 2005-2018 (pre-hype window); AI mentions classified actionable/speculative/irrelevant; causal design	panel econometrics	10-K text (actionable vs speculative vs irrelevant AI disclosures)
S29	Cao et al. (2025)	European Management Review (3)	USA	Study 1: instrument from 6,519 AI documents of 37 retailers; Study 2: fsQCA on survey of 140 U.S.-based retail managers	fsQCA	validated multi-level instrument (AI integration at task/function/firm level)
S32	D'Amico et al. (2025)	Journal of Technology Transfer (3)	UK	14,143 UK firms, 2004-2020	panel econometrics	archival: Beahurst-identified AI technology/product use, matched to Business Structure Database + UK Innovation Survey (2004-2020, biennial)
S34	Fontanelli et al. (2025)	Journal of Economic Behavior and Organization (3)	France	French ICT survey 2019, firm-level; balancing/matching robustness	panel econometrics	survey (predictive AI use; buyers vs in-house developers)
S01	Wamba-Taguimdje et al. (2020)	Business Process Management Journal (2)	multi-country	~500 vendor-published AI project mini-cases (IBM, AWS, Cloudera, Nvidia etc.), archival qualitative analysis	case study	case observation (AI-based transformation projects from vendor websites)
S10	Sun et al. (2022)	Systems Research and Behavioral Science (2)	China	234 AI-related manufacturers (GM/CEO respondents), CB-SEM	survey-SEM	survey construct (value co-creation in AI innovation ecosystem)

Table A.1 (continued)

ID	Study	Journal (AJG)	Country	Sample	Method	AI investment measure
S11	Ali et al. (2023)	Journal of Organizational Change Management (2)	UAE	single case (CMC Dubai): 9 semi-structured interviews with robotic-surgery team + archival data, triangulated	case study	case observation (adoption of AI/robotic surgery)

Table A.2: Included studies: outcomes, effect directions, and conditions (excerpt: 28 of 67 studies — the 25 in the highest-ranked journals (AJG 4*, 4, 3) plus the first three AJG 2 studies; within a rating by year). Source: author's compilation.

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S09	Mishra et al. (2022)	Perf.	Perf: gross and net operating efficiency, net profitability, ad-spend, employment	mixed		AI focus reduces gross operating efficiency but increases net operating efficiency and profitability - automation raises costs more slowly than revenues.
S14	Krakowski et al. (2023)	Both	Perf: competitive performance (game outcomes/ratings); CA: sources of persistent performance heterogeneity (= sustained advantage) across settings	conditional	substitution: traditional human capabilities become obsolete; complementation: new human-machine capabilities emerge that are unrelated or negatively related to traditional ones	AI adoption shifts the sources of competitive advantage - a new decision-making resource emerges at the human-AI intersection, unrelated to traditional capability.
S17	Babina et al. (2024)	Perf.	Perf: growth in sales, employment, and market valuation	positive	channel: product innovation rather than process efficiency; gains concentrate among firms that were already large, which raises industry concentration (superstar dynamics)	AI-investing firms grow in sales, employment and value via product innovation; benefits concentrate among large firms, increasing concentration.
S07	Leoni et al. (2022)	Perf.	Perf: perceived manufacturing firm performance (survey scale)	conditional	full mediation through knowledge management processes: the direct path from AI to manufacturing performance is not significant while both legs of the indirect path are; AI has no direct effect on supply chain resilience either; larger firms reach higher AI maturity	AI alone does not improve manufacturing firm performance - the effect exists exclusively through knowledge management processes (full mediation).
S45	Adams et al. (2026)	Perf.	Perf: growth in sales, employment, assets, markups; stock-return response to monetary policy	positive	adoption concentrated in larger, more productive firms; adopters more responsive to monetary policy surprises	Firms adopting AI-powered pricing grow faster in sales, employment and markups - AI pricing is a capability large productive firms exploit first.
S46	Agag et al. (2026)	Perf.	Perf: firm value; wage inequality as mediating channel	positive	wage inequality partially mediates: AI human capital compresses inequality and thereby raises firm value; the effect is stronger in dynamic and complex industries; by sector, the value effect is strongest in travel and the inequality compression in hospitality	AI human capital raises firm value partly by compressing wage inequality - a workforce-centered pathway to AI-enabled advantage, strongest in dynamic sectors.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S52	Filieri et al. (2026)	Perf.	Perf: supply chain efficiency, sales performance, market value	conditional	inverted U: AI performance benefits diminish at extreme tariff exposure ("optimal investment threshold"); customer engagement needs strengthen the effect (boundary condition); tariff exposure induces AI integration	Trade tensions push firms into AI integration that improves efficiency, sales and value - up to a threshold where adversity overwhelms the compensation.
S02	Baabdullah et al. (2021)	Perf.	Perf: perceived SME performance (survey scale)	positive	technology roadmapping (enabler, sig.); attitude (sig.); infrastructure readiness (sig.); awareness (sig.); professional expertise and technicality not significant	Acceptance of AI practices improves perceived SME performance; adoption is driven by roadmapping, attitude, infrastructure and awareness - not by professional expertise.
S03	Chatterjee et al. (2021)	Both	Perf: perceived organizational performance (survey scale); CA: competitive advantage scale (COA1-3, Rogers-based; incl. market-share gain), discriminant-valid vs OP; predicted by performance	positive	mediators: B2B engagement, employee experience, information processing capability (all sig.); moderator: leadership support (sig., MGA); firm size/age/industry only controls (not moderators)	AI-CRM implementation improves perceived organizational performance and competitive advantage in B2B relationship management.
S04	Chatterjee et al. (2022)	Perf.	Perf: firm performance scale (market+financial criteria, ROI/ROA, sales growth, market share, profitability) + B2B relationship satisfaction	positive	technology turbulence weakens the paths from automated decision-making and operational efficiency to B2B relationship satisfaction (negative moderator); leadership support strengthens the path from satisfaction to firm performance (positive moderator); the main effect of AI-CRM on firm performance is unconditional	AI-embedded CRM improves relationship satisfaction and firm performance; gains shrink under technology turbulence and grow with leadership support.
S05	Hossain et al. (2022)	CA	CA: sustained competitive advantage = relative performance vs competitors over last 3 years (market share growth, sales growth, profitability, ROI)	conditional	mediators: market sensing, seizing and reconfiguring carry the effect of marketing analytics capability on sustained competitive advantage in full; moderator: AI adoption strengthens all three capability paths	Marketing analytics capability drives sensing/seizing/reconfiguring toward sustained competitive advantage; AI adoption amplifies this only when built on an analytics foundation.
S06	Lee et al. (2022)	Perf.	Perf: revenue growth	conditional	AI-intensity threshold (low-level adoption: no effect); complementary technology investments (cloud, database) amplify; internal/exclusive R&D strategy amplifies	AI pays off only above an adoption-intensity threshold, and more so with complementary technology investments and venture-specific internal R&D.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S08	Lui et al. (2022)	Perf.	Perf: firm market value (CAR; -1.77% on event day)	negative	nonmanufacturing firms (more negative); weak IT capability (more negative); low credit rating (more negative)	investors react negatively to AI investment announcements on average; the penalty concentrates in nonmanufacturing firms with weak IT capability and low credit ratings.
S12	Czarnitzki et al. (2023)	Perf.	Perf: firm productivity (labor- and value-added-based)	positive		AI adoption is robustly associated with higher firm productivity in representative German data; the effect grows with usage intensity.
S15	Samadhiya et al. (2023)	Both	Perf: perceived firm performance (survey scale); CA: perceived sustainable competitiveness (survey scale)	positive	ICT capability (prerequisite); technological readiness (prerequisite); firm fit	AI-based partner relationship management improves firm performance and sustainable competitiveness, provided ICT capability and technological readiness are in place.
S16	Wu & Monfort (2023)	Perf.	Perf: perceived firm performance (survey scale)	positive	fsQCA configurations: marketing capabilities + customer value co-creation + market orientation + AI strategy jointly form necessary and sufficient recipes for high performance	AI marketing strategy raises firm performance, but only as part of configurations with marketing capabilities, co-creation and market orientation - a configurational result.
S18	Cannas et al. (2024)	Perf.	Perf: qualitative competitiveness outcomes: costs, lead times, service level, quality, safety, sustainability	positive	barriers as boundary conditions: data quality, AI skills, high investment needs, unclear economic benefits, lack of AI cost-analysis experience	AI improves supply-chain competitiveness across SCOR processes, but realization depends on overcoming data-quality, skill and investment-evaluation barriers.
S20	Pham et al. (2024)	Perf.	Perf: outpatient revenue, inpatient revenue, productivity, occupancy	positive	market share (larger-share hospitals adopt AI and benefit); endogeneity control essential - naive estimates misleading	Hospitals with larger market share adopt AI and gain in revenue, productivity and occupancy; without endogeneity controls the performance effects are misestimated.
S21	Sullivan & Fosso (2024)	Perf.	Perf: firm performance + process innovation + product innovation (survey scales)	conditional	adaptive response to market change is a full mediator, with no direct AI-to-performance paths modelled; environmental hostility weakens the automation path (negative moderator) and turns the relational path insignificant; environmental dynamism moderates as well; only ten of eighteen conditional indirect effects are significant	AI capabilities pay off through the firms adaptive response to market changes; environment hostility/dynamism condition how much each AI capability contributes.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S24	Alam et al. (2025)	Both	Perf: value creation (survey scale); CA: sustainable competitive advantage (survey scale, modeled as mediator)	positive	sustainable competitive advantage mediates between AI adoption and value creation; dynamic capabilities first impede value creation through transition costs and become crucial later under turbulence; technological turbulence moderates only the path from capabilities to value creation	AI adoption creates value in hospitality mainly through sustainable competitive advantage as transmission channel; dynamic capabilities help only once transition costs are absorbed.
S26	Banna & Alam (2025)	Perf.	Perf: firm revenue growth	conditional	U-shaped effect with a quantified turning point at roughly USD 11.3M of AI venture funding; below it AI drags revenue, above it AI pays; coupling AI with an R&D innovation strategy amplifies the effect, standalone AI is insufficient	AI investment follows a U-shaped payoff: short-term revenue drag, long-term gains - and only firms coupling AI with R&D strategy capture the upside.
S27	Basnet et al. (2025)	Perf.	Perf: firm value (market valuation); R&D spending, patents, lagged productivity as channels	conditional	what counts is the substance of the disclosure: actionable disclosures bring valuation gains, innovation and lagged productivity, speculative or irrelevant ones bring nothing, and silent or vague peers are penalised	Markets reward only substantive (actionable) AI disclosures - early adopters with real implementation plans gain value via innovation; vague AI talk earns nothing.
S29	Cao et al. (2025)	Perf.	Perf: efficiency and innovation outcomes (configurational)	conditional	configurational: no single AI function suffices - synergistic combinations (esp. customer service + cybersecurity with other functions); emergent capabilities: adaptive learning, predictive analytics, uncertainty mitigation	Retail performance comes from configurations of AI-enabled functions, not isolated AI uses - a directly configurational (fsQCA) result.
S32	D'Amico et al. (2025)	Perf.	Perf: innovation performance (knowledge spillover of innovation)	conditional	complementarity with own R&D investment (AI pays only if it complements internal R&D); AI substitutes knowledge collaboration with certain external partners	AI adoption boosts innovation only when paired with the firms own R&D investment - and partly replaces external knowledge collaboration.
S34	Fontanelli et al. (2025)	Perf.	Perf: volatility of productivity growth (not the level)	conditional	AI buyers: higher volatility; in-house developers: none; complementary human capital (share of ICT engineers/technicians) mitigates the buyer effect	Off-the-shelf AI raises the riskiness of productivity outcomes; developing in-house or holding complementary ICT human capital neutralizes it.
S01	Wamba-Taguimdje et al. (2020)	Perf.	Perf: organizational performance (financial, marketing, administrative) + process-level performance, qualitative	conditional	process reconfiguration required (mechanism: performance gains only when AI features are used to reconfigure processes); AI as configuration of complements (data, talent, domain knowledge, partnerships, infrastructure)	AI transformation projects improve organizational and process performance, but only when firms use AI to reconfigure their processes rather than overlay it on existing ones.

Table A.2 (continued)

ID	Study	Outcome	Outcome measure(s)	Direction	Conditions	Key finding
S10	Sun et al. (2022)	CA	CA: perceived competitive advantage scale (own construct, Table 3) + innovation intelligibility; no performance construct in model	conditional	full mediation: value co-creation has no direct effect on competitive advantage; dynamic capabilities and innovation capabilities carry the entire effect	Value co-creation in AI ecosystems has no direct effect on competitive advantage - effects run entirely through dynamic and innovation capabilities.
S11	Ali et al. (2023)	Both	Perf: four qualitative outcome categories: clinical (errors/infections down, patient experience), financial (quantified: ~500K/month at 30 robotic cases), organizational (resource allocation), technological; CA: competitive position with explicit competitor comparison: patient choice over competitors, bargaining power with third-party payers, differentiation, upper hand over competition (P1)	positive	qualified robotic team incl. revenue-cycle team (capability prerequisite); cultural shift in medical community; regulatory approval (DHA, case-by-case); high acquisition costs not covered by insurance (entry barrier)	AI-enabled robotic surgery improves clinical, financial and technological outcomes and strengthens the competitive position of the healthcare organization.